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Perspective

Synthetic tissue ecology: Ecological control of developmental self-organization across scales

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SUMMARY

Embryonic development builds tissues and organs through the self-organization of heterogeneous cells across spatial and temporal scales, analogous to an ecosystem. Although ecological systems are often stochastic and multistable, embryogenesis is remarkably reproducible. We argue that this contrast is not a contradiction. Developmental robustness arises from evolutionarily filtered ecological interactions embedded within hierarchically organized, multiscale architectures that restrict accessible collective states. Competition, cooperation, niche construction, and density-dependent coupling are not metaphors but core interaction dynamics that shape tissues *in vivo*. *In vitro* systems frequently lack these multiscale constraints, allowing more permissive ecological dynamics and increased variability. We propose "synthetic tissue ecology" as a conceptual and engineering framework that treats development as a stabilized regime of ecological organization. By shifting focus to interaction architectures and constraints that direct cross-scale self-organization, this framework enables the decoding of mesoscale modules and tissue-coupling principles while supporting the engineering of robust organoids, embryoids, and regenerative systems.

INTRODUCTION

Advances in stem cell biology, organoids, and synthetic developmental systems have enabled unprecedented access to human tissue formation *in vitro*.[1,2] Despite this progress, many engineered tissues remain fragile, variable, and difficult to reproduce. Small differences in initial conditions, geometry, or signaling context often lead to divergent outcomes, limiting both mechanistic insight and translational utility.[3–5] Translating single-cell phenotypes into predictable tissue- and organ-level behaviors therefore remains a major barrier to rational tissue engineering. These challenges suggest that prevailing design strategies, which emphasize cell-intrinsic specification and linear control of signaling pathways, capture only part of the logic governing multicellular tissue formation. A more complete engineering framework is needed that accounts for how target behaviors arise from interactions among cells.

Tissue organization emerges from interaction-dependent dynamics that couple cells to one another and to their environments across scales.[6,7] Embryonic development arises from coordinated behaviors among heterogeneous cells embedded in a responsive microenvironment. Cells compete for space, nutrients, and growth factors,[8,9] shaping survival and patterning, while cooperation among distinct lineages is essential for building functional tissues, echoing mutualistic relationships in ecological communities.[10] Cells also actively remodel their niches by depositing extracellular matrix (ECM), shaping morphogen gradients, and modifying mechanical and metabolic conditions that feedback on cell behavior.[11] Through these local interactions, embryos achieve large-scale coordination and robust self-organization without centralized control.[12]

In recent years, this interaction-centric view has been reinforced by efforts to build and interrogate multicellular systems *in vitro*. These studies suggest that, much as gene regulatory networks (GRNs) organize temporal programs of gene expression to generate cell fates, cell regulatory networks (CRNs) govern how cells collectively organize tissues and organs in space.[13] GRNs and CRNs are coupled through feedback involving mechanics, bioelectricity, geometry, and tissue-scale remodeling[14–16]: morphogenetic forces reshape signaling contexts and cell-cell interactions, which in turn feedback on gene expression, linking molecular programs to emergent tissue architecture.

How can multicellular systems be designed when function arises not from cell-intrinsic programs alone but from interaction-dependent dynamics operating across scales? Can we develop a unifying language capable of explaining cellular behavior at different levels of organization, supporting model construction, and guiding rational interventions? Here, we propose "synthetic tissue ecology" as a complementary framework for helping to understand and engineer multicellular systems. This framework treats tissues as populations of interacting cells whose collective behaviors emerge through competition, cooperation, feedback, and niche construction within

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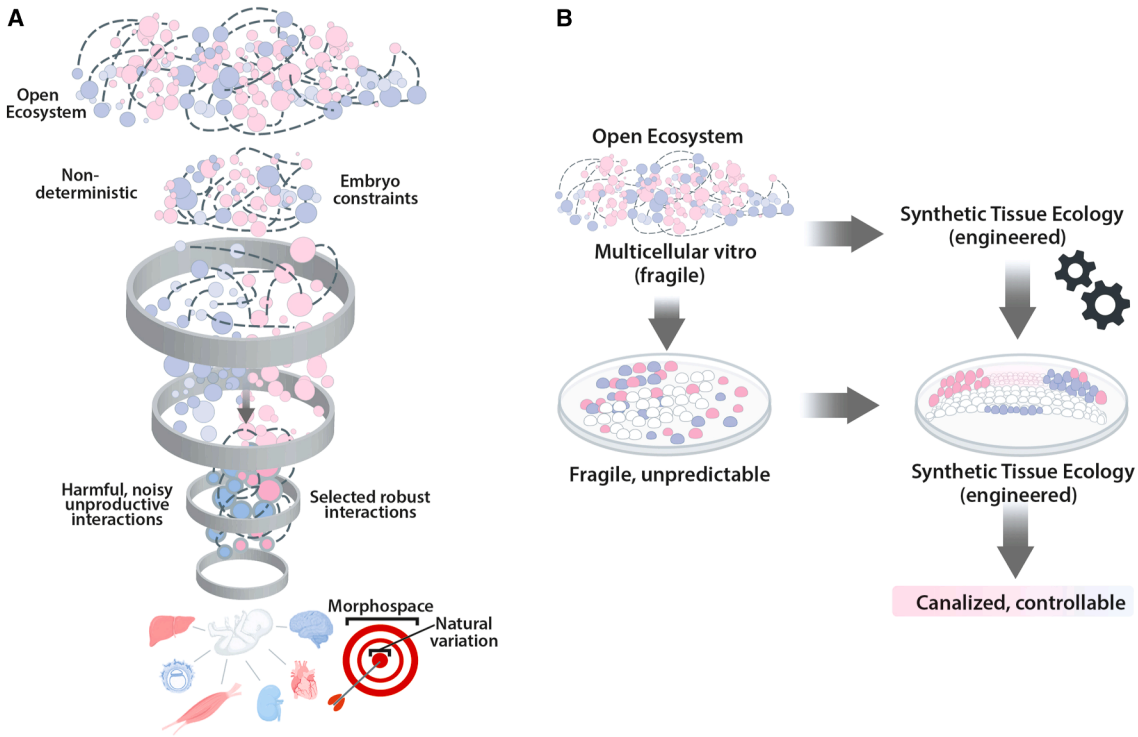


Figure 1. Synthetic tissue ecology canalizes multicellular organization by engineering interaction constraints

(A) Natural ecosystems explore a broad space of possible interaction networks and collective states, resulting in multiple potential outcomes. In embryogenesis, evolutionary selection progressively filters this interaction space, enforcing multiscale constraints that eliminate unstable configurations and retain only robust interaction architectures. This constraint-driven canalization restricts developmental morphospace and enables reproducible tissue- and organ-level outcomes despite local variability.

(B) *In vitro* multicellular systems often behave as open ecosystems with weakly constrained interactions, leading to fragile and unpredictable organization. Synthetic tissue ecology proposes that robustness can be restored by engineering interaction architectures so that collective behavior becomes canalized into stable and controllable developmental regimes.

Adobe Illustrator was used to make the schematic.

constrained environments. Here, regulation does not act by directly specifying outcomes, but instead, it limits the range of possible collective behaviors by constraining interactions across scales. We do not argue that embryos directly mirror natural ecosystems. Instead, we use ecological principles as an engineering language that aids understanding of organizing motifs and that can be implemented and tested in multicellular systems.

In the sections that follow, we reconcile ecological dynamics with developmental determinism and outline how ecological principles can guide the rational design and generation of more robust and reproducible synthetic tissues.

DEVELOPMENT AS A CONSTRAINED ECOLOGICAL SYSTEM

At first glance, ecological framing appears to conflict with a defining feature of embryogenesis: its remarkable determinism and reproducibility. Natural ecosystems are often stochastic and capable of occupying multiple stable states, whereas embryos reliably produce a single outcome. We argue that this contrast reveals the core insight of an ecological perspective instead of undermining it. Developmental determinism is not achieved by eliminating ecological interactions but by constraining them. Embryogenesis represents a highly restricted regime of ecological organization in which evolution has filtered the space of possible interactions, retaining only those that reliably construct a viable organism[17–20] (Figure 1A).

From this perspective, development can be understood as a post-evolutionary, domesticated ecosystem. Evolution acts as the exploratory phase, sampling diverse organizational motifs through competition, cooperation, and niche construction across deep time. Development then executes a single, stabilized configuration of these motifs. All embryonic cells share a common genome and exist within a closed population, enforcing shared interaction rules while allowing context-dependent behaviors to arise from differences in state, position, and local environment. Developmental hierarchies and feedback networks enforce these evolutionarily derived constraints, compressing ecological dynamics into a narrow, reproducible state by suppressing alternative trajectories. This reduction in ecological degrees of freedom underlies the robustness of embryogenesis. Additionally, the constraints promote coupling of developmental events within one scale or across scales, further enabling self-organization and emergence.

SYNTHETIC TISSUE ECOLOGY AS A DESIGN FRAMEWORK

Current approaches in tissue engineering and synthetic biology often borrow metaphors from electrical and mechanical

engineering, emphasizing modularity, standardization, and deterministic control.[21] For example, differentiation protocols typically focus on specifying individual lineages in isolation, and orthogonal patterning strategies are used to impose vascular or structural organization at the cellular level. While those designs that treat tissues as traditional machines can be effective for addressing certain biological questions, imposing extreme control over cells limits the ability to mimic complex living behaviors that depend on context, feedback, and emergent structure.[22] Many *in vitro* tissues are unstable and difficult to reproduce, not because developmental programs are inherently fragile but because essential interactions between cells and their physical, mechanical, and potentially electrochemical environments are absent or weakened[23–26] (Figure 1B).

Evidence for this view spans biological systems: microbial communities converge to reproducible states under defined spatial and resource limitations,[27–29] and human pluripotent stem cell systems display robust patterning only when geometric, metabolic, or boundary conditions are imposed, as seen in micropatterned colonies and confined gastruloids[30–34] (comprehensively reviewed in Cao et al.[22]). These observations suggest that reproducibility emerges less from control at the level of individual components and more from structuring interactions within constrained environments. As such, developmental robustness emerges because ecological interactions are not isolated at the cellular level but embedded within a multilevel architecture in which molecular rules, cell behaviors, tissue modules, and global geometry reciprocally constrain one another. Yet we currently lack a general design language for constructing, perturbing, and comparing interaction architectures, one that treats interaction type, timing, and coupling strength as core engineering variables.

In ecology, decades of theoretical work have provided formal tools for predicting how interacting populations assemble, maintain stability, and transition under environmental and interaction constraints. Synthetic tissue ecology extends this interaction-centered logic to developmental systems by formalizing multicellular organization in terms of structured interactions and the constraints that shape them. Within this framework, interaction topology, coupling strength, spatial organization, and resource availability define the design space, reframing engineering from individual cell types to the interaction networks that organize them.

Practically, this enables two complementary strategies: constructing minimal interacting systems to systematically probe how local interaction rules scale to tissue-level organization, and selectively reconstructing key interaction motifs, such as mechanical coupling, competition, feedback, or niche construction, to restore robustness *in vitro* (Figure 1B). Beyond engineering, this approach provides a principled route for discovering mesoscale modules that bridge single-cell behavior and organ-scale structure. By systematically varying interaction parameters, synthetic tissue ecology allows the identification of recurrent population-level and community-level motifs, including boundary zones, source-sink territories, feedback loops, and mechanically coordinated tissue regions that stabilize and organize collective dynamics. These mesoscale modules reveal how hierarchical self-organization emerges through cross-scale coupling, where microscale interactions give rise to cellular behaviors that assemble into structured tissue architectures. In this way, synthetic tissue ecology serves not only as a design framework but also as a discovery platform for decoding the principles that govern multilevel organization in living systems.

ECOLOGICAL INTERACTIONS ACROSS DEVELOPMENT, DISEASE, AND STEM CELL SYSTEMS: LESSONS FROM EXISTING STUDIES

Although ecological reasoning is rarely invoked in developmental biology or tissue engineering, many studies implicitly show that multicellular behavior is governed by population-level interactions. For example, neighborhood analysis suggested that progenitor cell decisions are driven by context rather than absolute autonomous cell programs.[35,36] Recent work has begun to formalize tissues in terms of mesoscale modules, intermediate "supracellular" organizational units that bridge single cells and organs and are defined by patterned cell-cell communication, spatial coupling, and collective dynamics.[6,7,37] Interaction-aware analyses of spatial transcriptomics have revealed reproducible tissue modules that cannot be inferred from cell states alone but instead emerge from local signaling architecture and neighborhood structure.[38]

Conceptual frameworks further argue that such modules represent fundamental units of tissue function, providing a missing link between molecular programs and organ-scale behavior. A recent study aimed to simplify how we model complex multicellular systems by creating a "cell behavior grammar" that translates plain language into computer simulations of entire cellular ecosystems. This is vital because understanding how diverse cells interact and evolve together as a multicellular unit is essential for predicting outcomes. By testing this framework, the authors demonstrated that it can accurately digitize complex multicellular processes, such as how cancer cells invade tissues, how immune cells coordinate to attack tumors, and how the brain organizes its layers during development.[39]

In ecological systems, analogous mesoscale structures such as patches, colonies, or aggregates emerge when local interactions are amplified by environmental coupling, allowing populations to stabilize collective states without centralized control.[40] From this perspective, supracellular domains function as ecological units whose persistence reflects interaction topology, feedback, and environmental context rather than the properties of individual agents. Their emergence illustrates how local interactions give rise to reproducible higher-order organization. Two recent studies further show that tissue patterning emerges from reciprocal interactions between cells and the supracellular structures they generate. By framing tissue as an ecosystem, skeleton patterning was shown to emerge from structural bistability within the mesenchyme, driven by cross-scale interactions between cell states and supracellular structures akin to adaptive ecological organization, enabling self-compartmentalization into cartilage and soft tissue fates.[41] In parallel, morphogens such as fibroblast growth factor (FGF) and bone morphogenetic protein (BMP) have been shown to act by inducing distinct supracellular physical states such as solid-like cores and contractile fluid margins that interact to drive tissue deformation and organ budding (e.g., skin).[42] Mechanically driven neighborhood organization has also been observed in systems such as soft-matrix-induced

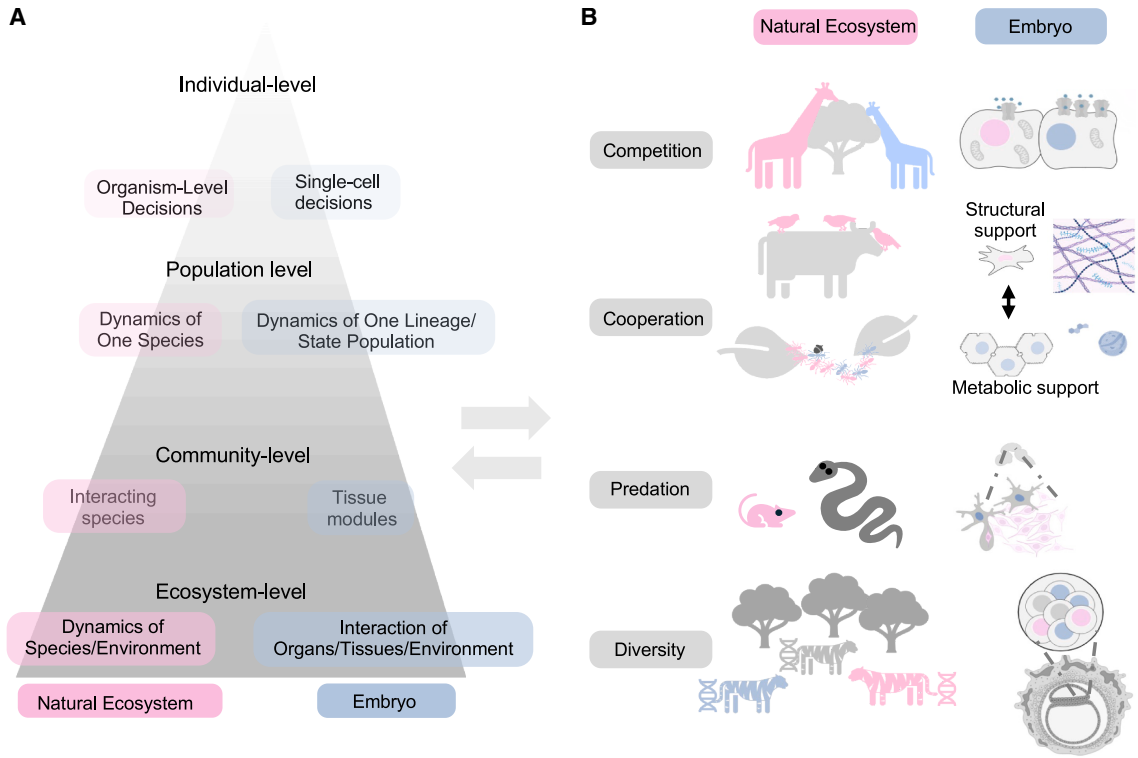


Figure 2. Ecological concept in embryogenesis

(A) Hierarchy of organization in natural ecosystems and embryos. Ecological analysis spans individual-, population-, community-, and ecosystem-level dynamics. An analogous hierarchy can be applied to embryogenesis, from single-cell decisions to lineage population dynamics, interacting tissue modules, and organism-scale coupling between tissues and environment.

(B) Cross-scale interaction motif analogies between ecosystems and embryos. Key ecological interaction classes such as competition, cooperation, predation-like selective clearance, and diversity have clear developmental counterparts that shape multicellular organization and robustness.

BioRender and PowerPoint were used to make the schematics.

monocyte clustering and mesenchymal condensation during early development.[43–45] Together, these studies demonstrate that tissue patterning and morphogenesis emerge from cross-scale interactions between cells and supracellular structures, with signaling molecules acting by shaping collective physical states rather than directly specifying outcomes.

Manipulating interaction structures can confer robustness that is difficult to achieve through molecular specification alone. For example, while simple negative feedback can regulate population size, such systems are highly susceptible to "cheater" mutations that evade control. By contrast, paradoxical circuit architecture couples a single quorum-sensing signal to both growth promotion and inhibition, creating an intrinsic self-limiting control system that is resistant to cheater escape. This design produces a biphasic survival response, in which proliferation is permitted only within a defined signal range. By linking survival and programmed death to the same signal, the circuit ensures that mutants unable to sense the signal are eliminated by the death-inducing arm. Experimental implementations in mammalian cells demonstrate that these architectures can maintain stable population homeostasis for extended periods (up to 42 days), effectively counter-selecting sensing-deficient variants without requiring detailed control over internal cellular states.[46]

Minimal mammalian multicellular systems further reinforce the importance of ecologically inspired views and constrained interaction architectures. A well-characterized example is the macrophage-fibroblast exchange circuit, in which reciprocal paracrine signaling between two cell types forms a stable, homeostatic minimal system. Importantly, stability arises from specific interaction features, including environmental constraint, negative feedback, and contact-dependent coupling.[47] Related principles operate in engineered developmental systems, where synthetic organizer populations stabilize morphogen gradients and improve the reproducibility of symmetry breaking without requiring precise specification of internal cell states.[48]

Cancer biology provides a contrasting example of these same principles operating in a deregulated regime. Tumors behave as evolving ecosystems composed of interacting cellular populations that compete for space and nutrients, cooperate through shared growth factors, and actively remodel their microenvironment.[49–52] Framing tumors as ecological systems has enabled predictive therapeutic strategies, such as adaptive therapy, that exploit population dynamics instead of aiming for total eradication.[53–55] Developing tissues rely on similar interaction dynamics but operate within strong constraints imposed by shared genetic identity, spatial boundaries,

Box 1. Early development through an ecological lens

Early embryonic development, and gastrulation in particular, illustrates development as a hierarchically organized ecological process.[62] Within a narrow window, gastrulation integrates cell-level variability, population-scale competition and coordination, reciprocal interactions among distinct tissues, and embryo-wide state transitions into a regulated developmental program.[63] Germ layer formation and primitive streak morphogenesis arise not from a linear genetic script alone but from interactions among heterogeneous cells, competition for limiting inductive cues, cooperation between embryonic and extraembryonic tissues, and feedback-driven global reorganization.[64]

At the level of individual cells, heterogeneity functions as a developmental resource, not merely biological noise. During early stages, cells display variable transcriptional states and signaling responsiveness, yet these differences remain reversible and probabilistic.[65] This variability may distribute developmental competence across the population, preventing synchronized commitment and helping embryos to buffer fluctuations in signaling or cell number. After implantation, differences in chromatin state, metabolism, and cell cycle phase diversify cellular thresholds for responding to patterning cues, contributing to spatial organization without requiring perfectly precise morphogen gradients. This strategy closely parallels ecological response diversity, in which phenotypic variation stabilizes populations by spreading risk across individuals.[66]

At the population scale, gastrulation is shaped by competition, resource limitation, and collective behavior. Embryonic cells compete for access to space, metabolic resources, and morphogens, and developmental outcomes reflect relative fitness and not absolute metrics. Cells with reduced competence can be selectively eliminated when juxtaposed with fitter neighbors, biasing lineage contributions and protecting tissue integrity in a manner analogous to competitive exclusion in ecological communities.[59,67] Spatial constraints and embryo geometry impose an additional layer of regulation: tissue mechanics, such as local cell confinement and tension, influences primitive streak formation and morphogenesis. Within this framework, the timing and position of epiblast cells determine whether they ingress to form mesoderm or endoderm or remain in the epiblast to acquire ectoderm fates.[68] Superimposed on these competitive dynamics are collective behaviors, including coordinated tissue flows and transitions between solid-like and fluid-like states.[69–71] Mathematical frameworks such as reaction-diffusion, cellular automata, active matter, and collective computation capture these population-level transitions and provide a shared language for understanding how local interactions give rise to emergent tissue organization.[72–75]

At the community level, gastrulation depends on reciprocal interactions among embryonic and extraembryonic tissues that can be framed as mutualism and niche construction. Extraembryonic tissues shape signaling, transport, and mechanics required for embryonic patterning, while embryonic tissues provide inductive and positional cues needed for extraembryonic organization.[76–79] These reciprocal dependencies stabilize developmental trajectories.

At the ecosystem level, gastrulation exhibits disturbance, succession-like progression, feedback regulation, and resilience. For instance, the progression from pre-streak epiblast to organizer emergence resembles ecological succession, in which early states modify conditions to enable later ones.[80]

EXAMPLES OF ECOLOGICAL MOTIFS IN DEVELOPMENT

Succession-like transitions in gastrulation

Gastrulation can be interpreted as a succession-like sequence of state transitions in which earlier cellular rearrangements generate the conditions required for later patterning. BMP, Nodal, and WNT signaling initiate gastrulation-associated programs, through which epiblast cells acquire lineage identities and spatial coordinates. Ingressing primitive-streak derivatives then remodels tissue geometry and mechanics to create new spatial relationships between signaling centers and responding tissues.[81] As in ecological succession, each stage modifies the environment in ways that constrain and enable the next.

Predation-like clearance as quality control

Development also contains predation-like quality control processes that remove inappropriate or excess cellular states while preserving system-level stability. Microglia-mediated elimination of neural precursors and synaptic pruning provide clear examples of selective clearance shaping tissue architecture.[82,83] More broadly, apoptosis, phagocytic clearance, and selective organelle elimination act as culling modules that refine population composition. Unlike predator-prey systems, however, these processes are evolutionarily constrained and do not generate open-ended population oscillations, highlighting how developmental determinism canalizes predation-like interactions.

Niche partitioning through resource and signal competition

Niche partitioning can emerge during development through competition for spatially restricted signals and metabolic resources, together with differential access to these cues. During gastrulation, gradients of morphogens such as Nodal define localized regions of high signaling activity in which only a subset of cells reaches induction thresholds, while others experience weaker exposure and adopt alternative fates.[84] Metabolic constraints further shape fate competence and lineage allocation, as metabolic pathways and intermediates influence signaling, epigenetic state, and differentiation potential.[5,85–88] Together, spatial access, timing, and resource availability create partitioned developmental trajectories that resemble resource-driven niche differentiation.

(Continued on next page)

Box 1. *Continued*

Together, gastrulation illustrates how embryonic development can be seen as a constrained ecological process in which multiscale organizations emerge from competition, cooperation, niche construction, feedback, and succession. This ecological lens links individual variability to embryo-scale order and provides guiding principles for engineering self-organizing multicellular systems.

lineage coupling, and multiscale feedback, which channel ecological interactions.[56,57]

Quantitative studies of cell competition provide strong evidence that fate acquisition is a population-level process shaped by cell-cell competition and relative fitness as opposed to a uniform, cell-autonomous response. Using cellular barcoding, lineage tracing, and mathematical modeling, Shakiba, Zandstra, and colleagues demonstrated that reprogramming to induced pluripotency is governed by competitive clonal selection, in which a small subset of high-fitness "elite" clones disproportionately expand and ultimately define the resulting population despite identical starting genotypes and culture conditions.[58,59] Recent work has extended interaction-centric principles to complex human embryoid models using evolutionary game theory. By modeling behaviors such as proliferation and cell killing as conditional strategies whose payoffs depend on population size and collective survival, these studies show that actions costly to individual cells can improve outcomes at the level of the embryo. In this framework, the same behavior can shift from harmful to beneficial depending on collective state, becoming active only when continued growth increases the risk of failure.[60] In parallel, Frenster et al. used mosaic gastruloids to show that developmental cell competition is temporally restricted, with p53-deficient supercompetitor cells eliminating neighboring wild-type cells during a defined window between primed pluripotency and early gastrulation.[61]

Together, these studies demonstrate that across development, disease, and stem cell systems, robust multicellular organization emerges from structured and constrained interaction architectures, highlighting the need for a unifying framework that treats interaction regimes as primary drivers of collective cellular dynamics.

These parallels can be organized across scales. Individual cells, lineage populations, tissue modules, and whole embryo-environment interactions correspond conceptually to individual-, population-, community-, and ecosystem-level organization in natural ecosystems (Figure 2A). Across these scales, competition, cooperation, predation-like interactions, and diversity can shape collective organization in both ecosystems and developing tissues (Figure 2B). To illustrate how these interaction principles operate across scales within a concrete developmental context, we apply this ecological framework to gastrulation in Box 1.

ECOLOGICAL THINKING TOWARD MULTICELLULAR SYSTEM ENGINEERING

What does ecological thinking add to stem cell multicellular engineering beyond traditional biological or engineering approaches? We argue that its value is 3-fold:

First, ecology can offer a unifying language for interaction-driven systems. Concepts such as competition, coexistence, succession, and carrying capacity help explain why multicellular assemblies stabilize, reorganize, or collapse in ways that are difficult to capture through molecular signaling alone. This helps place diverse studies within a shared conceptual framework. *Second*, ecological thinking enables hypothesis generation that lineage-centric reasoning often overlooks, including whether the absence of partner populations destabilizes fate, whether unchecked competition drives clonal dominance, or whether insufficient heterogeneity underlies *in vitro* irreproducibility. *Third*, it offers a practical blueprint for synthetic development by emphasizing how interaction strength, timing, and spatial organization constrain cross-scale outcomes.

For instance, ecology emphasizes that habitats are history-dependent: early inhabitants can modify an environment in ways that persist, shaping where and how later species can settle (priority and legacy effects).[89,90] Viewing developmental systems through this lens suggests that early mesodermal populations can deposit ECM and trap secreted ligands within it, leaving behind a long-lasting biochemical and mechanical imprint. Later-arriving cells do not encounter a neutral space, but a pre-conditioned microenvironment that biases their signaling exposure, positioning, and fate decisions. In another example, ecological thinking holds that niches are actively constructed, not simply occupied. For instance, beavers do not merely inhabit environments but actively reshape them by building dams that generate entirely new ecosystems. Similarly, migrating cells can be viewed as niche engineers rather than passive migrants that simply adapt to a pre-existing microenvironment. For instance, primordial germ cells transiently occupy the yolk sac/allantoic region, and hematopoietic stem cells (HSCs) transiently colonize the fetal liver during development. These migratory or colonizing progenitor populations may remodel the transient tissue environments they encounter, helping establish supportive niches rather than simply adapting to pre-existing ones. This thinking style changes the engineer's or biologist's view of the problem.

This interaction-centric thinking and design is not speculative. It has shown effectiveness in synthetic microbial ecology, where robust multicellular systems are engineered by sculpting collective dynamics instead of optimizing individual organisms.[91,92] In microbial consortia, uncontrolled competition often leads to instability, whereas programming interdependence through mutualism or shared resource constraints stabilizes community behavior.[93,94] Spatial structuring further constrains interactions, enabling division of labor and emergent functions that fail in well-mixed environments.[29,95,96]

ECOLOGICAL DESIGN LEVERS FOR ENGINEERING

Ecological interactions can be deliberately introduced or tuned using *in vitro* synthetic biology tools, such as engineered signaling circuits, adhesion modules, inducible feedback, and

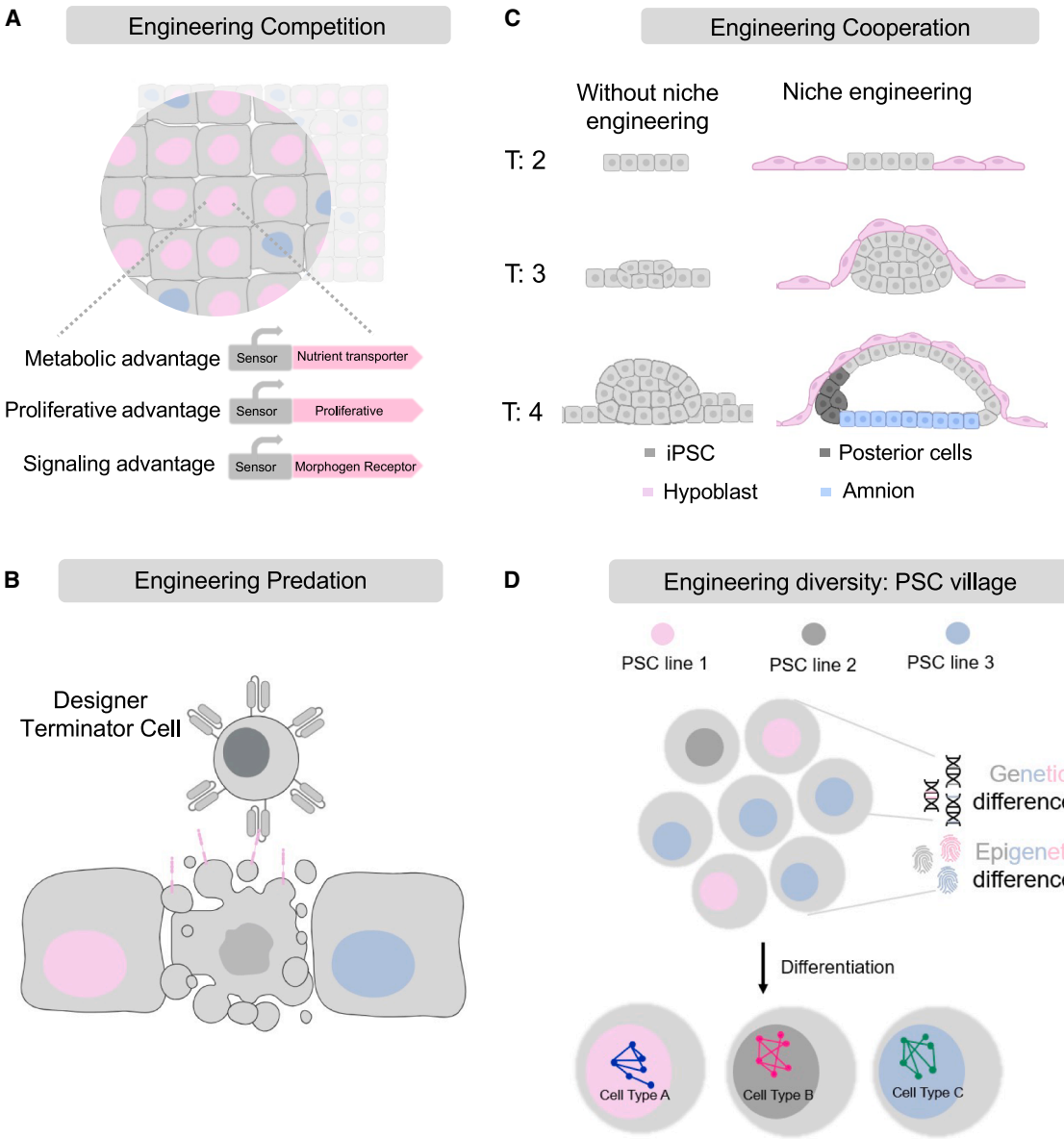


Figure 3. Synthetic tissue ecology, integrating ecology, and synthetic biology for engineering tissue morphogenesis

(A) Engineering synthetic competition. Cells with an engineered competitive advantage (pink) are shown outcompeting other cells in coculture. This advantage can be introduced using synthetic biology to provide a metabolic edge (e.g., enhanced nutrient uptake), a proliferative edge (e.g., overexpressing cell-cycle proteins), or a signaling edge (e.g., heightened responsiveness to morphogens).

(B) Targeted cell elimination through synthetic predation. Engineered "predator" cells, like chimeric antigen receptor (CAR)-T cells or macrophages, are programmed to recognize a specific surface antigen on "prey" cells. This recognition triggers the targeted destruction and removal of the unwanted cell population from the culture.

(C) Cooperative interactions enable tissue self-organization. This example highlights niche construction within *heX-embryoids*.[97] On the left, induced pluripotent stem cells (iPSCs) cultured alone fail to organize, forming a disordered, piled-up layer. On the right, coculture with engineered hypoblast-like cells provides a crucial niche, guiding the iPSCs to self-assemble into higher-order structures mimicking early embryonic development, including the amniotic cavity and posterior pole.

(D) Engineering diversity in culture. The heterogeneity of a cell culture can be increased by mixing pluripotent stem cells derived from different individuals. Inherent genetic and epigenetic differences between the cell lines result in distinct differentiation biases, leading to greater diversity in the final tissue.

BioRender and PowerPoint were used to make the schematics.

programmable cell elimination. Competition shapes tissue composition by biasing lineage expansion through differential access to nutrients, space, and morphogen responsiveness (Figure 3A). Predation-like clearance provides an additional layer of quality control, enabling selective removal of maladaptive, mispatterned, or undifferentiated cells (Figure 3B). Cooperation and niche construction enable higher-order organization through lineage-specific provision of ECM, chemical, electrical, and mechanical cues that stabilize tissue-scale architecture (Figure 3C). Diversity further builds developmental trajectories by distributing

Table 1. Actionable design principles for synthetic tissue ecology

Ecological design verb	Developmental analog	Interaction-level design levers	Expected mesoscale or system-level outcome
Compete	lineage competition for space, time, nutrients, and morphogens	control nutrient access, adhesion strength, crowding, signal sensitivity, or fitness-linked circuits (e.g., survival coupling and Myc-like selection)	selective expansion of desired populations; pruning of maladaptive or off-target cells
Cooperate	reciprocal signaling between embryonic and niche-forming lineages	co-development of complementary populations; engineered paracrine signaling; ligand-presenting cells or beads (e.g., Notch[98] and FGF)	establishment of lineage identity and coordinated pattern formation
Construct niches	ECM deposition, morphogen shaping, and mechanical conditioning	engineered matrix production, boundary formation, localized signal sources, mechanical confinement or patterning	context-dependent fate specification and spatial organization
Maintain diversity	heterogeneous progenitor competence and lineage bias	mixed genetic or epigenetic cell populations organized into interacting neighborhoods (e.g., iPSC "villages" and state-diverse progenitors)[99,100]	robustness, modular specialization, and adaptive capacity at the mesoscale
Cull/prune (controlled clearance)	apoptosis and quality control during morphogenesis	inducible cell death,[101] eliminator populations, density- or stress-triggered kill switches, optogenetic ablation[102]	maintenance of tissue size, composition, and developmental quality
Self-limit (feedback and carrying capacity)	density-dependent growth and fate decisions	density-sensing circuits,[46] growth-limiting feedback, resource restriction	preservation of tissue architecture and prevention of overgrowth

functional roles across heterogeneous populations (Figure 3D). Together, these principles illustrate how ecological design can guide self-organization by restoring interaction structures that impose constraints and establish collective outcomes across scales. Table 1 summarizes actionable design principles, and Box 2 presents three case studies.

CHALLENGES AND PERSPECTIVES

Ecologically inspired multicellular engineering provides a conceptual framework for understanding and engineering multicellular systems. Because it draws on established ecological concepts with deep analytical and experimental foundations, this approach advances the broader goal of building a unifying language for probing tissue self-organization. Ecology is already a mature multiscale science of living systems, developed to explain how local interactions scale into population and landscape-level structure. It offers a functional vocabulary and structured classification of interaction motifs. Although elements of this approach have appeared individually in prior studies and can carry semantic interpretations, its true power lies in its organization within a coherent framework that specifies how multiple types of interaction couple and constrain one another and how they collectively stabilize or reorganize multicellular architecture.

Synthetic tissue ecology does not replace established approaches in engineering development, such as morphogen-gradient engineering, biomaterial-based spatial patterning, or modular genetic-circuit design. In fact, this framework provides a complementary population-level perspective that organizes how these tools are combined, coupled, and constrained over time to shape collective behavior. In this view, gradients, materials, and circuits specify local rules, while ecological design governs how those rules interact to produce stable and reproducible tissue-scale outcomes. This perspective may also enhance deterministic pluripotent stem cell differentiation by clarifying principles of niche interaction and temporal constraint that improve lineage efficiency, potency, and functional maturity.

Stable mesoscale tissue modules, defined by recurring interactions and spatial coupling, can be functional without resembling actual organs.[6] These intermediate units bridge single-cell states and tissue-level organization and can be identified and engineered. Advances in single-cell profiling, spatial genomics, and inferred communication networks have dramatically expanded our ability to catalog cellular states and potential interactions, but they have not produced a corresponding ability to predict or control tissue-level behavior. This gap reflects a conceptual mismatch between how interactions are measured and how collective organization emerges. Current single-cell communication frameworks often infer who can signal to whom, implicitly assuming that communication potential alone explains organization. In practice, whether interactions remain local or become system-spanning depends on ecological parameters such as interaction strength, density, spatial reach, feedback, and boundary conditions. Without embedding communication within these parameters, inferred networks struggle to predict symmetry breaking, coordinated morphogenesis, or collective state transitions. Some ecologically inspired approaches are beginning to appear in spatial tissue analysis, as exemplified by recent work that adapts ecological diversity and heterogeneity metrics to quantify spatial organization and neighborhood complexity in tissue-scale data.[114]

This limitation reflects a broader cross-scale problem in how complex systems organize. Classic ecological regime shifts, such as lake eutrophication,[115] population collapse driven by Allee effects,[116] or vegetation-mediated desertification, illustrate how gradual changes in local interactions can trigger abrupt

Box 2. Case studies: Ecological design for building multicellular systems *in vitro*

The case studies below focus on execution, but the framework emphasizes that diagnosis is equally important. Together, they illustrate how this perspective reframes common embryoid failure modes as experimentally tractable questions about interaction architecture, population-level decision-making, and developmental regime shifts. An ideal ecological engineering strategy adopts design-build-test iterative cycles in which multimodal data collected across space and time are integrated with computational models to map interaction structures and collective states, and targeted interventions are used to reconstruct or modulate missing interactions. Repeating these cycles progressively refines interaction architectures, builds necessary missing constraints, and improves target outcomes (e.g., form and function).

It is important to note that the goal is not to build a fully *in vivo*-like system capturing every interaction, which would create an impractically large and unmanageable engineering space. Instead, engineering should be driven by specific questions and defined problems. For example, why cell-type composition is not robust or why the efficiency of lumenogenesis remains low.

CASE STUDY 1: DELAYED COMMITMENT EMBRYOIDS AND COLLECTIVE TIMING THRESHOLDS

A central challenge in current stem cell engineering is the reliance on predefined morphogen exposure schedules, which assume that all multicellular assemblies progress at a uniform rate. However, *in vitro* systems are inherently variable, and small differences in initial cell numbers or geometry can lead to divergent outcomes when differentiation is ''forced'' by external signaling cocktails. This case study proposes an ecological shift: treating developmental timing as an emergent, population-dependent variable rather than an externally imposed script.

The problem

Traditional protocols often follow a linear sequence, adding ''factor X'' on a specific day, regardless of the system's internal state. This ignores the fact that developmental transitions *in vivo* often function as thresholded regime shifts, where local interactions must reach a certain density or frequency before they can trigger a system-wide state change. Forcing commitment before a population has reached the necessary carrying capacity or spatial organization leads to the ''fragile'' and ''variable'' results common in current embryoid models.

The ecological pivot

In ecological systems, state transitions are frequently gated by critical thresholds in density, resource availability, or stress. To mirror this, progenitor cells can be engineered with delayed competence windows or density-sensing circuits (e.g., quorum-sensing). Under this design, lineage commitment programs are only activated once internally sensed collective conditions such as a specific cellular density or mechanical stress level are met. (1) *Collective feedback*: use synthetic circuits that link fate transition to population-level feedback. This ensures that the transition only occurs when the population is large enough to support robust tissue architecture. (2) *Self-correction*: by gating transitions through collective thresholds, the system corrects for differences in initial cell numbers, preventing early variability from propagating into divergent outcomes. (3) *Self-timed response*: timing can also be stabilized at an individual cell level using self-timing circuits in which engineered sensors allow reading cell state and trigger actuation only upon reaching a defined competence threshold. By coupling morphogen sensitivity to internal state instead of external time, these circuits reduce dependence on rigid schedules and buffer variation between embryoids.

This framework allows to test whether developmental failure *in vitro* stems from incorrect patterning cues or simply from mistimed commitment. In this design, the system will determine its own transition point based on its collective state. This approach may show reproducibility in synthetic tissues from structuring interactions within constrained environments that allow collective timing to emerge predictably.

CASE STUDY 2: ECOLOGICALLY INSPIRED STRATEGY FOR A REPRODUCIBLE ANTERIOR-POSTERIOR AXIS

A common failure mode in embryoid models is low efficiency in forming a polarized CER1+ anterior domain opposite a TBXT+ posterior pole.[103] Although CER1-mediated antagonism of Nodal signaling can help spatially restrict TBXT induction, CER1 patterning is often unstable *in vitro*, appearing either too sparse to generate a coherent inhibitory domain or overly abundant and poorly regionalized, flattening spatial information and disrupting axis formation.

Synthetic tissue ecology reframes this as a coupled population problem involving Allee-effect-like thresholds, where collective function emerges only once a population exceeds a critical density or connectivity.[104] Specifically, the CER1+ cell population may need to cross an effective density threshold to transition from scattered individual inhibitors into a coherent inhibitory territory. Once established, this CER1 inhibitory domain imposes a second threshold on TBXT expression in neighboring cells by restricting where posteriorizing signaling inputs can rise above the level required for TBXT+ posterior domain formation.

(Continued on next page)

Box 2. *Continued*

Problem

This interaction resembles spatial ecological models in which one population restricts the distribution of another through inhibitory fields or interference competition. Examples include allelopathy in plants or bacteriocin-producing bacterial strains that carve out spatial domains. These dynamics can be formalized using spatial competition models and activator-inhibitor frameworks, where the balance between short-range activation and long-range inhibition determines whether stable polarized domains emerge.

The ecological pivot

The ecological framing suggests engineering strategies to increase axis formation efficiency. *First*, engineer the CER1-expressing anterior domain to reliably cross its effective density threshold by controlling initial aggregate size, local clustering, and coupling range. In ecology, populations often aggregate around resource hotspots such as lakes or watering holes. Analogously, embryoids can be given an engineered "lake" that recruits and concentrates CER1+ cells. This can be implemented by micropatterned or localized release of CER1-inducing/cell-recruiting cues or by embedding a localized reservoir in the matrix or microfluidic delivery to create a stable anterior high-attractant zone. CER1+ cells can be engineered to express an inducible, orthogonal adhesion module[105,106] so that only CER1+ cells preferentially adhere to one another and maintain a contiguous anterior domain. *Second*, stabilize regionalization by introducing interaction rules that promote CER1 self-limitation outside the anterior while maintaining strong coupling within the anterior domain. For example, synthetic Notch circuits can be used to make CER1 identity conditional on local neighborhood context so that CER1 is reinforced by homotypic contact and extinguished when CER1+ cells stray into incompatible neighborhoods.[107,108] *Third*, prevent ectopic CER1-expressing cell expansion using selective filtering mechanisms that mirror ecological territorial exclusion. By eliminating misplaced CER1+ cells through position- or contact-dependent control, the system enforces spatial boundaries and maintains robust patterning. *Finally*, adjust the strength and diffusion range of inhibitory versus activating signals so that TBXT is not eliminated everywhere, but instead forms a stable TBXT+ posterior domain opposite the CER1+ anterior region. For instance, this may be done by tuning how much CER1 is produced or how long its effect persists outside cells.

Importantly, the overall goal is not to micromanage fates but to canalize a two-population interaction architecture into a stable spatial configuration where an anterior inhibitory territory forms reliably and a posterior TBXT+ domain emerges as its constrained complement. In most cases, not all of these implicated layers are defective. One or two targeted interventions is usually sufficient to increase the frequency of appropriately formed tissues. Therefore, an exhaustive, multi-layered approach should ideally be avoided.

CASE STUDY 3: ENGINEERING ACROSS SCALES BY DESIGNING BOUNDARIES

In ecology, boundaries are functional habitats rather than passive lines. Ecotones, the interface zones between ecological communities, concentrate competition, exclusion, and information flow, enabling stable segregation while permitting controlled exchange. For example, forest-grassland edge supports species from both systems, plus edge specialists. Some species, such as white-tailed deer, feed in open grass and shelter in the forest. Predators also learn that boundaries are high-traffic zones.[109]

In tissues, boundaries arise from microscale rules such as polarized adhesion, contact-dependent inhibition, or interface-specific survival circuits. These local interaction rules generate spatially confined boundary habitats that resist mixing and regulate signal propagation. When properly specified, such boundaries canalize morphogenesis by preventing fate interpenetration and preserving compartment identity and mechanical features (reviewed comprehensively in Martyn and Gartner).[110] In some cases, the combined signal environment produces a new multipotent lineage not present in either territory alone (i.e., neural crest cells emerge at the boundary between neural plate and non-neural ectoderm[111]).

In epiblast-extraembryonic endoderm embryoids, the proper formation of a boundary between the two cell populations is required for the final pro-amniotic cavity formation. A stable interface enables deposition of laminin layer by hypoblast-like cells, resulting in polarization of wild-type epiblast cells, emergence of microlumen and expansion toward the cavity.[112,113] By contrast, loose or unstable boundaries fail to induce effective cavitation, demonstrating that boundary integrity is not merely architectural but also instructive.

Future research focused on understanding how boundaries direct self-organizing behavior will clarify their utility as engineering control points. Within this framework, microscale interaction rules determine how cells sense and respond; mesoscale modules, such as ecotones, define population geography; and tissue-scale organization emerges when these modules collectively constrain expansion, feedback, and mechanical coupling toward certain developmental outcomes. Cross-scale design, therefore, consists of engineering local interaction parameters to produce mesoscale territories that contribute to tissue-scale developmental programs.

transitions into new collective states governed by different effective rules.[117] In these systems, organization does not increase smoothly with incremental changes in individual behavior. Instead, once a critical threshold in interaction strength, density, or coupling is crossed, local processes suddenly percolate across the system, leading to rapid reorganization.

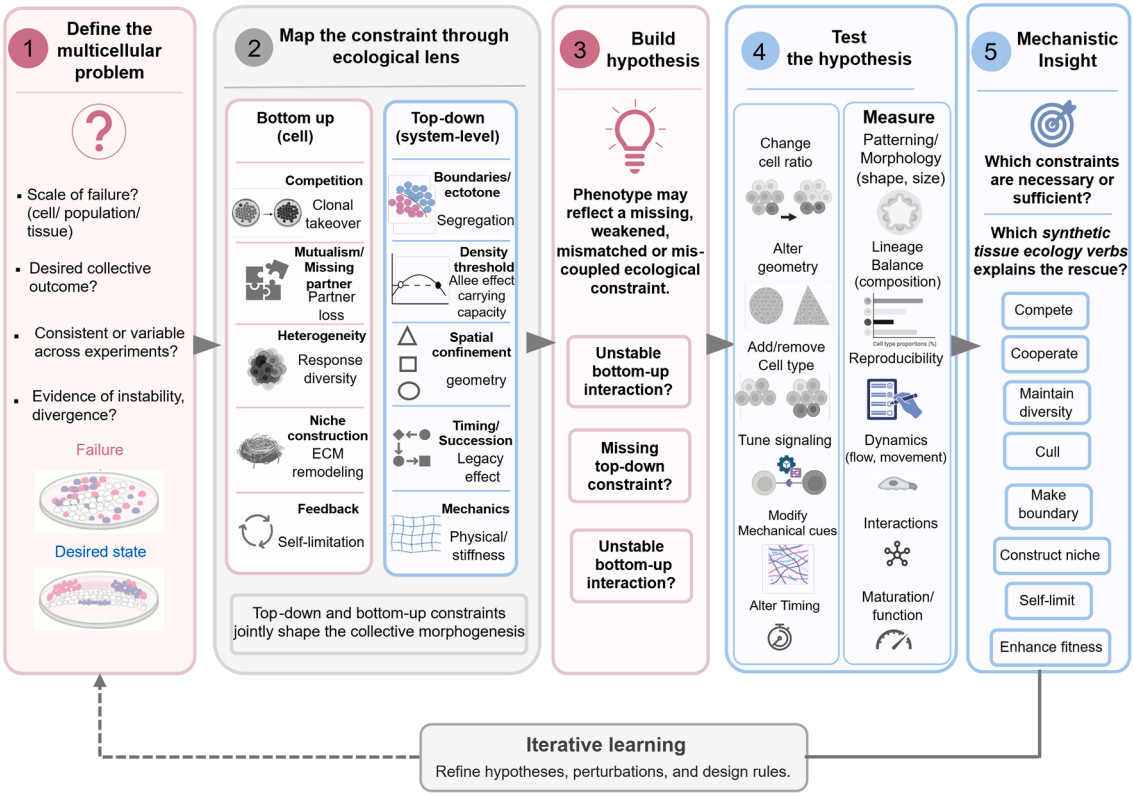


Figure 4. A roadmap for ecology-inspired synthetic developmental biology

Synthetic tissue ecology provides a framework for converting multicellular phenotypes into testable system-level hypotheses. The roadmap begins by defining the problem, including the scale of failure, desired collective outcome, reproducibility across experiments, and evidence of instability or divergence. The phenotype is then interpreted through an ecological lens by mapping bottom-up cell-level interactions together with top-down system-level constraints. These coupled constraints jointly shape collective morphogenesis and define the accessible states of the system. The resulting diagnosis is used to formulate hypotheses in which the phenotype reflects a missing, weakened, mismatched, or mis-coupled ecological constraint. These hypotheses can be tested by perturbing variables including, but not limited to, cell ratios, geometry, cell-type composition, signaling, mechanical cues, or timing, and by measuring patterning, lineage balance, reproducibility, dynamics, interactions, maturation, and function. The roadmap further frames the resulting design logic through synthetic tissue ecology verbs, including compete, cooperate, enhance fitness, cull, make boundary, construct niche, self-limit, and maintain diversity. Iterative testing identifies the constraints that are necessary or sufficient for the desired collective state and extracts generalizable design rules for engineering multicellular systems. BioRender and PowerPoint were used to make the figure.

Developmental systems (e.g., gastrulation) follow a similar logic. Organization does not increase smoothly from cells to tissue-level patterns. Instead, developmental progression occurs through threshold-dependent state transition, in which synergies that were previously local are amplified, become collective, and generate new system-level states.[118,119] Below these thresholds, perturbations remain confined and decay. Once crossed, the system operates under new effective rules that cannot be inferred by averaging lower-scale behaviors. To this end, a further challenge is deciding when and how to intervene. Many engineering strategies rely on continuous control, yet ecological systems reorganize only when key parameters such as density, interaction range, coupling strength, or resource availability cross critical thresholds. Effective intervention therefore depends less on imposing outcomes and more on identifying moments when systems are poised for reorganization. Agent-based models, cellular Potts models, and nonlinear dynamical systems can encode interaction rules, spatial constraints, and feedback, making them well-suited for exploring and identifying regime boundaries in multicellular systems.[120–124] As systems approach these boundaries, they exhibit predictive signatures such as increased variance or correlation length, indicating when small perturbations can have disproportionate effects. Systems biology modeling and future AI/machine learning approaches can therefore support the design, testing, and implementation of interaction architectures by helping map collective state space, identify boundaries, and predict minimal interventions that can trigger stable reorganization. Emerging multiscale foundation model efforts, such as AI-driven digital organism (AIDO)-based approaches[125] and spatially informed models such as Nicheformer,[126] suggest a shift toward learning tissue-level biology, where cell states are interpreted within their spatial context to infer interaction strengths between cell populations. In this view, AI helps to learn high-dimensional dynamics that can underlie emergent tissue behavior. Coupled with perturbation datasets and active-learning experimental design, these approaches may enable large-scale *in silico* exploration of interaction space, identify minimal constraint architectures, and prioritize interventions to channel stem cells into reproducible, functional tissue forms.

Ecological thinking also clarifies why heterogeneity often promotes developmental robustness instead of causing instability.[127] Recent gastruloid studies support this view by showing that pre-existing clonal diversity can promote robust axial organization through division of labor among biased stem cell clones, suggesting that heterogeneity can serve as a constructive population-level feature rather than simply a source of variability.[128] Persistent differences in lineage competence or regulatory state enable reciprocal interactions that stabilize higher-order organization. Similar ideas have emerged in cell manufacturing, where heterogeneity is increasingly viewed as a population-level feature that enhances robustness and scalability.[129] In multicellular tissue and embryo-model systems, early heterogeneity, including variable expression of lineage-associated regulators such as GATA-binding factor 6 (GATA6), can support cooperative assembly of complex architectures through shared modification of signaling, mechanical, and metabolic environments.[97] From an ecological perspective, such heterogeneity enables exploration of interaction space and access to collective regimes that are inaccessible under uniform or overly constrained conditions.

Development is robust not because outcomes are specified in detail but because interaction dynamics are constrained to allow reproducible transitions from cells to tissues and organs. This perspective provides a practical roadmap for hypothesis generation and problem solving in multicellular systems (Figure 4). Starting from an observed multicellular phenotype, the roadmap asks which ecological constraints may be missing, weakened, mistimed, or mis-coupled across top-down system-level features and bottom-up cellular interactions and then links this diagnosis to targeted perturbations and cross-scale readouts. By identifying and engineering the ecological parameters that gate these transitions, we can move beyond eliminating variability and instead design contexts in which interaction-driven self-organization and collective behavior emerge reliably. Synthetic tissue ecology is not a final solution or an endpoint. It opens new conceptual and experimental directions toward next-generation human-based multicellular system engineering, an approach that will mature through iterative practice.

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AUTHOR CONTRIBUTIONS

K.K. and M.R.E. wrote the original text of the manuscript. K.K. made the figures, and M.R.E. was involved in providing input to the final version of it.

DECLARATION OF INTERESTS

M.R.E. and K.K. have pending patent applications related to multicellular systems. M.R.E. is a co-founder of Genexgen Inc., Orgagen Biosciences Inc., and heXembio Inc. and also acts as a member of their scientific advisory board or management. The authors affirm that this research was conducted independently and that they have no affiliations or financial interests that could be construed as potential conflicts of interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used Google Gemini, ChatGPT, and Anthropic Claude for proofreading of the text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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